

Experiment 5

Aim - To Build the pipeline of jobs using Maven / Gradle / Ant in Jenkins, create a pipeline script to Test and deploy an application over the tomcat server

Programming in Jenkins:

Continuous Integration is a software development practice where members of a team integrate their

work frequently, usually each person integrates at least daily leading to multiple integrations per day.

Each integration is verified by an automated build (including test) to detect integration errors as

quickly as possible.” In simple way, Continuous integration (CI) is the practice of frequently building

and testing each change done to your code automatically.

Jenkins is a self-contained, open-source automation server which can be used to automate all sorts of

tasks related to building, testing, and delivering or deploying software.

Our first job will execute the shell commands. The freestyle project provides enough options and

features to build the complex jobs that you will need in your projects.

Example 1

Example 1.1: Deploying a freestyle app in Jenkins

Creating a job:

Naming the job and setting it as freestyle:

Selecting build type as “Execute shell”:

Entering a simple command for the shell execution:

Applying and saving the project configuration:

Building the project:

Console output (after building):

Example 1.2: Taking parameters through files

Contents of script example1.cmd:

Executing script example1.cmd on the terminal:

Modifying the Jenkins project to execute the script while supplying required parameters:

Console output after building the modified project:

Example 2

Example 2.1: Running a Java program under Jenkins

Creating a simple Java program:

Compiling and running the program on the terminal:

Creating a new freestyle project:

Configure new project:

Console output after building:

Example 3

Example 3.1: Parameterise build

Creating a new freestyle project:

Enabling parameterisation and adding a String parameter:

Configuring the string parameter as Fname:

Adding a choice parameter and configuring it as City with the following choices:

Creating a script which takes 2 arguments for name and city:

Configuring build steps:

Entering parameters for build:

Console output after building:

Example 3.2: Running a Java program with parameters

Creating a Java program with an input argument:

Testing the program on the terminal:

Creating a new freestyle project:

Parameterise the project by adding a string parameter as follows:

Configure the build steps:

Entering the parameter for the build:

Console output after building:

Example 5

Example 5.1: Running a Python program

Creating a simple Python script:

Running the Python script on the terminal:

Creating a new freestyle project:

Parameterising the project with a string parameter as follows:

Configuring the build steps:

Setting the parameter for the build:

Console output after building:

Conclusion: Thus, we have successfully studied Continuous Integration and installed, configured, and understood programming with Jenkins.

The image shows two screenshots of the Jenkins web interface. The top screenshot is the Jenkins dashboard at localhost:8080. It features a 'Welcome to Jenkins!' message and a 'Start building your software project' section with buttons for 'Create a job', 'Set up an agent', 'Configure a cloud', and 'Learn more about distributed builds'. The bottom screenshot is the 'New Item' page, where a new job named 'piggy bank' is being created. The 'Pipeline' item type is selected. The page includes a search bar, a list of item types (Freestyle project, Multi-configuration project, Folder, Multibranch Pipeline), and an 'OK' button at the bottom. The browser's taskbar at the bottom shows the system date and time as 13:34 on 27-01-2026.

Jenkins Dashboard (localhost:8080)

REST API Jenkins 2.528.3

New Item

Enter an item name

piggy bank

Select an item type

Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

Folder
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.

Multibranch Pipeline
Creates a set of Pipeline projects according to detected branches in one SCM repository.

OK

The image displays two screenshots of the Jenkins web interface, showing the configuration page for a job named "piggy bank".

Top Screenshot: The "General" tab is selected under the "Configure" menu. The "General" section shows the job is "Enabled". The "Description" field is empty. Below it, there are several checkboxes: "Discard old builds", "GitHub project", "This project is parameterized", "Throttle builds", and "Execute concurrent builds if necessary", all of which are currently unchecked. An "Advanced" dropdown is visible. The "Source Code Management" section shows "None" selected for the SCM provider. At the bottom, there are "Save" and "Apply" buttons.

Bottom Screenshot: The "Build Steps" tab is selected under the "Configure" menu. The "Command" section shows a text area containing the command `echo "Hello Tsec"`. Below the text area is an "Advanced" dropdown and a "+ Add build step" button. The "Post-build Actions" section shows a "+ Add post-build action" button. At the bottom, there are "Save" and "Apply" buttons. A green "Saved" notification bar is visible at the bottom left of the interface.

The screenshot shows the Jenkins Configuration page for a job named 'piggy bank'. The left sidebar contains a menu with 'General', 'Source Code Management', 'Triggers', 'Environment' (selected), 'Build Steps', and 'Post-build Actions'. The main area is titled 'Configure' and has several checkboxes: 'Delete workspace before build starts', 'Use secret text(s) or file(s)', 'Add timestamps to the Console Output', 'Inspect build log for published build scans', 'Terminate a build if it's stuck', and 'With Ant'. Below these is the 'Build Steps' section, which includes a description and a list of build steps. One step is 'Execute Windows batch command', which has a command field containing 'echo Hello %TSC%'. A green 'Saved' message is visible at the bottom left, and 'Save' and 'Apply' buttons are at the bottom right.

The screenshot shows the Jenkins Job page for the 'piggy bank' job. The left sidebar contains a menu with 'Status' (selected), 'Changes', 'Workspace', 'Build Now', 'Configure', 'Delete Project', and 'Rename'. The main area is titled 'piggy bank' and has a green status icon. Below the title is a 'Permalinks' section with a list of build links: 'Last build (#13), 2 sec ago', 'Last stable build (#13), 2 sec ago', 'Last successful build (#13), 2 sec ago', 'Last failed build (#12), 1 min 32 sec ago', 'Last unsuccessful build (#12), 1 min 32 sec ago', and 'Last completed build (#13), 2 sec ago'. Below the permalinks is a 'Builds' section with a search filter and a list of builds. The builds are listed with their status (green for success, red for failure), build number, and time. The builds are: #13 (1:45 PM), #12 (1:43 PM), #11 (1:42 PM), #10 (1:42 PM), #9 (1:41 PM), #8 (1:40 PM), and #7 (1:40 PM). A green 'Saved' message is visible at the bottom left, and 'Save' and 'Apply' buttons are at the bottom right.

The screenshot shows a web browser window displaying the Jenkins interface. The address bar shows the URL `localhost:8080/job/piggy%20bank/lastBuild/console`. The Jenkins logo and navigation links are visible on the left. The main content area shows the 'Console Output' for a build named 'piggy bank #13'. The output text is as follows:

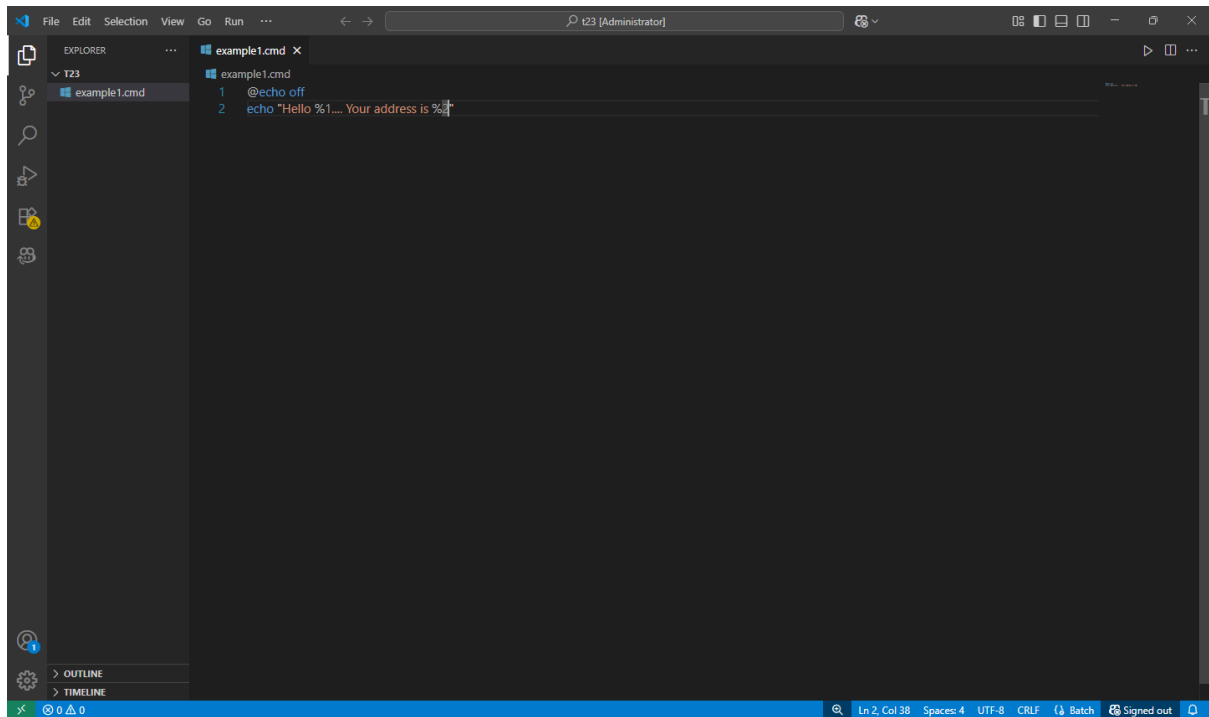
```
Started by user Tija Nathwani
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\piggy bank
[piggy bank] $ cmd /c call C:\WINDOWS\TEMP\jenkins898928918577088250.bat

C:\ProgramData\Jenkins\jenkins\workspace\piggy bank>echo Hello Tsec
Hello Tsec

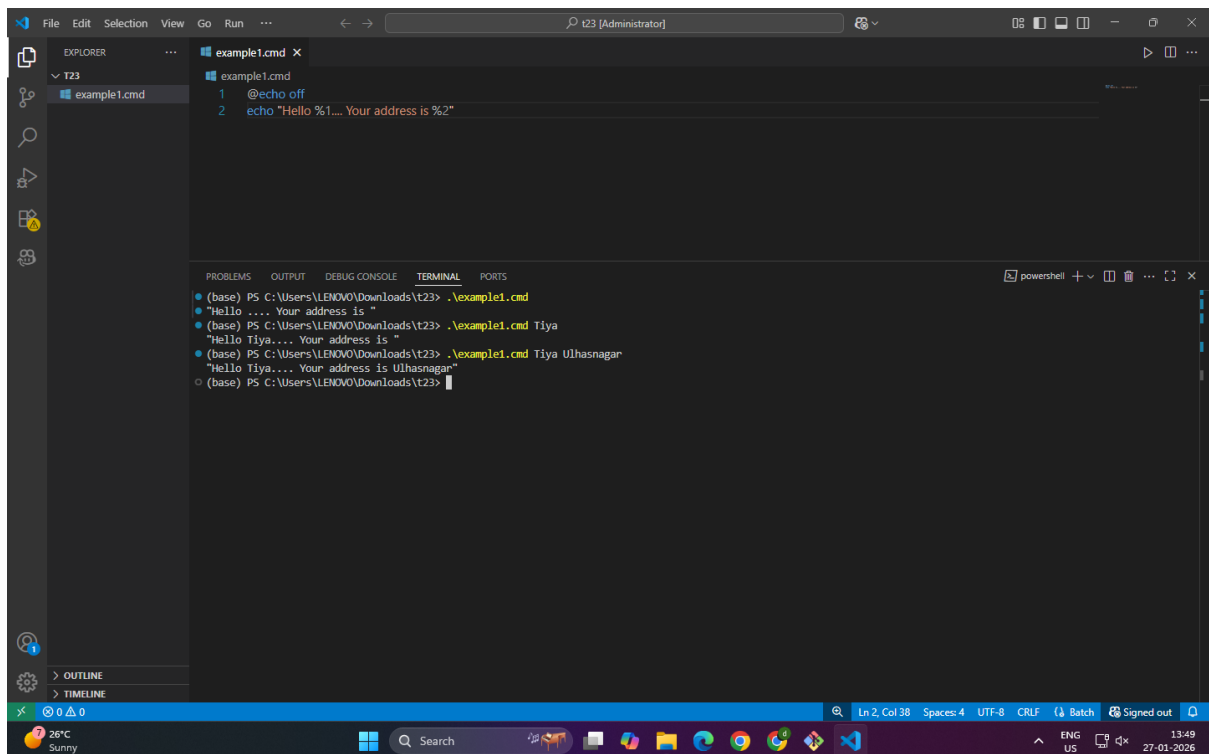
C:\ProgramData\Jenkins\jenkins\workspace\piggy bank>exit 0
Finished: SUCCESS
```

At the bottom of the browser window, the Windows taskbar is visible, showing the Start button, search bar, and various application icons. The system tray on the right shows the date and time as 13:45 on 27-01-2026.

1.2



```
File Edit Selection View Go Run ... t23 [Administrator]
EXPLORER
  t23
    example1.cmd
example1.cmd
1 @echo off
2 echo "Hello %1.... Your address is %2"
```



```
File Edit Selection View Go Run ... t23 [Administrator]
EXPLORER
  t23
    example1.cmd
example1.cmd
1 @echo off
2 echo "Hello %1.... Your address is %2"

TERMINAL
powerShell
• (base) PS C:\Users\LENOVO\Downloads\t23> .\example1.cmd
• "Hello .... Your address is "
• (base) PS C:\Users\LENOVO\Downloads\t23> .\example1.cmd Tiya
• "Hello Tiya.... Your address is "
• (base) PS C:\Users\LENOVO\Downloads\t23> .\example1.cmd Tiya Ulhasnagar
• "Hello Tiya.... Your address is Ulhasnagar"
• (base) PS C:\Users\LENOVO\Downloads\t23>
```

The screenshot shows the Jenkins Configuration page for a job named 'piggy bank'. The 'Environment' tab is selected in the left sidebar. The main area contains a list of checkboxes for environment configuration: 'Delete workspace before build starts', 'Use secret text(s) or file(s)', 'Add timestamps to the Console Output', 'Inspect build log for published build scans', 'Terminate a build if it's stuck', and 'With Ant'. Below these is the 'Build Steps' section, which includes a step titled 'Execute Windows batch command'. The command field contains the text 'C:\Users\LENOVO\Downloads\123\example1.cmd Ttya Ullhasnagar'. A green 'Saved' message is displayed at the bottom left of the configuration area. The browser's address bar shows 'localhost:8080/job/piggy%20bank/configure'.

This screenshot is identical to the one above, showing the Jenkins Configuration page for the 'piggy bank' job. However, the status at the bottom left of the configuration area has changed to a green bar with the text 'Build Now Done.' instead of 'Saved'. The rest of the interface, including the configuration options and the 'Execute Windows batch command' step, remains the same.

The screenshot shows a web browser window displaying the Jenkins interface. The address bar shows the URL `localhost:8080/job/piggy%20bank/lastBuild/console`. The Jenkins logo and navigation links are visible on the left. The main content area shows the 'Console Output' for a build named 'piggy bank'.

Console Output

```
Started by user Tiya Nathwani
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\piggy bank
[piggy bank] $ cmd /c call C:\WINDOWS\TEMP\jenkins15919224875493028413.bat

C:\ProgramData\Jenkins\jenkins\workspace\piggy bank>C:\Users\LENOVO\Downloads\t23\example1.cmd Tiya Uhasnagar
"Hello Tiya.... Your address is Uhasnagar"
Finished: SUCCESS
```

The bottom of the image shows a Windows taskbar with the date and time 13:51 on 27-01-2026, and the system tray showing the temperature as 26°C and the weather as Sunny.

Example 2

The image shows a screenshot of a web browser displaying the Jenkins configuration page for a job named 'Example 2'. The browser tabs include 'Java Downloads | Oracle', 'SEPM - Google Drive', 'Experiment 5 - Google Docs', '(38) WhatsApp', and 'Example 2 Config - Jenkins'. The address bar shows 'localhost:8080/job/Example%202/configure'. The Jenkins page has a left sidebar with 'Configure' selected, showing options like 'General', 'Source Code Management', 'Triggers', 'Environment', 'Build Steps', and 'Post-build Actions'. The 'General' tab is active, showing a 'Description' field, a 'Plain text' preview, and several checkboxes: 'Discard old builds', 'GitHub project', 'This project is parameterized', 'Throttle builds', and 'Execute concurrent builds if necessary'. The 'Source Code Management' section is set to 'None'. Below the configuration page, the VS Code editor is open, showing a file explorer with 'example1.cmd', 'Example2.class', 'Example2.java', and 'Main.class'. The editor displays the code for 'Example2.java', which is a simple Java class with a 'main' method that prints 'Hello World!'. The terminal at the bottom shows the command prompt '(base) PS C:\Users\LENOVO\Downloads\t23>'.

Jenkins Configuration Page (Example 2):

- General tab selected.
- Description field is empty.
- Plain text preview is empty.
- Discard old builds: ☐
- GitHub project: ☐
- This project is parameterized: ☐
- Throttle builds: ☐
- Execute concurrent builds if necessary: ☐
- Source Code Management: None

VS Code Editor:

```
1 class Example2 {
2     public static void main(String args[]) {
3         System.out.println("Hello World!");
4     }
5 }
```

Terminal:

```
(base) PS C:\Users\LENOVO\Downloads\t23>
```

The image shows two screenshots of the Jenkins web interface. The top screenshot displays the 'Configure' page for a job named 'Example 2'. The left sidebar contains a menu with 'General', 'Source Code Management', 'Triggers', 'Environment', 'Build Steps', and 'Post-build Actions'. The 'Environment' tab is selected. On the right, there are several checkboxes: 'Delete workspace before build starts', 'Use secret text(s) or file(s)', 'Add timestamps to the Console Output', 'Inspect build log for published build scans', 'Terminate a build if it's stuck', and 'With Ant'. Below these is the 'Build Steps' section, which includes a step named 'Execute Windows batch command'. The command field contains the text 'C:\Users\LENOVO\Downloads\123\Example2.java'. The bottom screenshot shows the 'Example 2' job page. The left sidebar has 'Status', 'Changes', 'Workspace', 'Build Now', 'Configure', 'Delete Project', and 'Rename'. The main area shows 'Example 2' with a 'Permalinks' section. Below this, there is a 'Builds' section showing a single build with a status icon and a blue and white striped progress bar. At the bottom, a green notification bar states 'Build scheduled'. The bottom status bar shows 'REST API Jenkins 2.528.3' and a system tray with a clock and language settings.

localhost:8080/job/Example%202/configure

Jenkins / Example 2 / Configuration

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute Windows batch command

Command

See the list of available environment variables

C:\Users\LENOVO\Downloads\123\Example2.java

Advanced

Save Apply

localhost:8080/job/Example%202/

Jenkins / Example 2

Status

Changes

Workspace

Build Now

Configure

Delete Project

Rename

Example 2

Permalinks

Add description

Builds

Today

#1 2:00 PM

Build scheduled

localhost:8080/job/Example 2/build?delay=0sec

REST API Jenkins 2.528.3

The screenshot shows a web browser window with multiple tabs. The active tab is titled 'Example 2 #1 Console - Jenkins'. The address bar shows the URL 'localhost:8080/job/Example%202/1/console'. The Jenkins interface is displayed, with a sidebar on the left containing links for 'Status', 'Changes', 'Console Output' (which is selected), 'Edit Build Information', and 'Timings'. The main area is titled 'Console Output' and shows a progress bar. Below the progress bar, the console output text is visible: 'Started by user Tija Nathwani', 'Running as SYSTEM', 'Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\Example 2', '[Example 2] \$ cmd /c call C:\WINDOWS\TEMP\jenkins14948875696471621075.bat', and 'C:\ProgramData\Jenkins\jenkins\workspace\Example 2>C:\Users\LENOVO\Downloads\t23\Example2.java'. At the bottom of the browser window, the Windows taskbar is visible, showing the 'Nifty bank' widget, the search bar, and various application icons. The system clock in the bottom right corner indicates the time as 14:01 on 27-01-2026.

Java Downloads | Oracle x SEPM - Google Drive x Experiment 5 - Google Docs x Example 2 #1 Console - Jenkins x +

localhost:8080/job/Example%202/1/console

Jenkins / Example 2 / #1 / Console Output

Status
Changes
Console Output
Edit Build Information
Timings

Console Output

Progress: [Progress Bar]

Download Copy View as plain text

Started by user [Tija Nathwani](#)
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\Example 2
[Example 2] \$ cmd /c call C:\WINDOWS\TEMP\jenkins14948875696471621075.bat
C:\ProgramData\Jenkins\jenkins\workspace\Example 2>C:\Users\LENOVO\Downloads\t23\Example2.java

REST API Jenkins 2.528.3

Nifty bank
14:01
27-01-2026

Example 3

The image displays two screenshots of the Jenkins web interface, illustrating the setup of a new job.

Top Screenshot: New Item Page

- The browser address bar shows `localhost:8080/view/all/newJob`.
- The page title is "New Item".
- The "Enter an item name" field contains "Example 3".
- The "Select an item type" section shows several options:
 - Pipeline**: Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
 - Freestyle project**: Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications. (This option is highlighted with a light blue background).
 - Multi-configuration project**: Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
 - Folder**: Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
 - Multibranch Pipeline**: Creates a set of Pipeline projects according to detected branches in one SCM repository.
- An "OK" button is visible at the bottom.

Bottom Screenshot: Configuration Page

- The browser address bar shows `localhost:8080/job/Example%203/configure`.
- The page title is "Configure".
- The left sidebar shows the configuration menu with "General" selected.
- The main content area shows the "General" tab with the following settings:
 - ☐ GitHub project
 - ☒ This project is parameterized ?
 - ☐ Add Parameter (dropdown menu is open showing options: Boolean Parameter, Choice Parameter, Credentials Parameter, File Parameter, Multi-line String Parameter, Password Parameter, Run Parameter, String Parameter)
 - ☐ Filter
 - ☐ Boolean Parameter
 - ☐ Choice Parameter
 - ☐ Credentials Parameter
 - ☐ File Parameter
 - ☐ Multi-line String Parameter
 - ☐ Password Parameter
 - ☐ Run Parameter
 - ☐ String Parameter
- Buttons for "Save" and "Apply" are at the bottom.

The image displays two screenshots of the Jenkins configuration page for a job named 'Example 3'. The top screenshot shows the 'String Parameter' configuration, and the bottom screenshot shows the 'Choice Parameter' configuration.

Top Screenshot: String Parameter Configuration

- Configuration Page:** Jenkins / Example 3 / Configuration
- Left Sidebar:** General (selected), Source Code Management, Triggers, Environment, Build Steps, Post-build Actions.
- String Parameter Configuration:**
 - Name:** fname
 - Default Value:** (empty)
 - Description:** (empty)
 - Trim the string:** ☐
- Buttons:** Save, Apply

Bottom Screenshot: Choice Parameter Configuration

- Configuration Page:** Jenkins / Example 3 / Configuration
- Left Sidebar:** General (selected), Source Code Management, Triggers, Environment, Build Steps, Post-build Actions.
- Choice Parameter Configuration:**
 - Name:** City
 - Choices:** Bandra, Kalyan, Mulund
 - Description:** (empty)
- Buttons:** Save, Apply

The image shows a Jenkins configuration page for a job named 'Example 3' and a Visual Studio Code editor window.

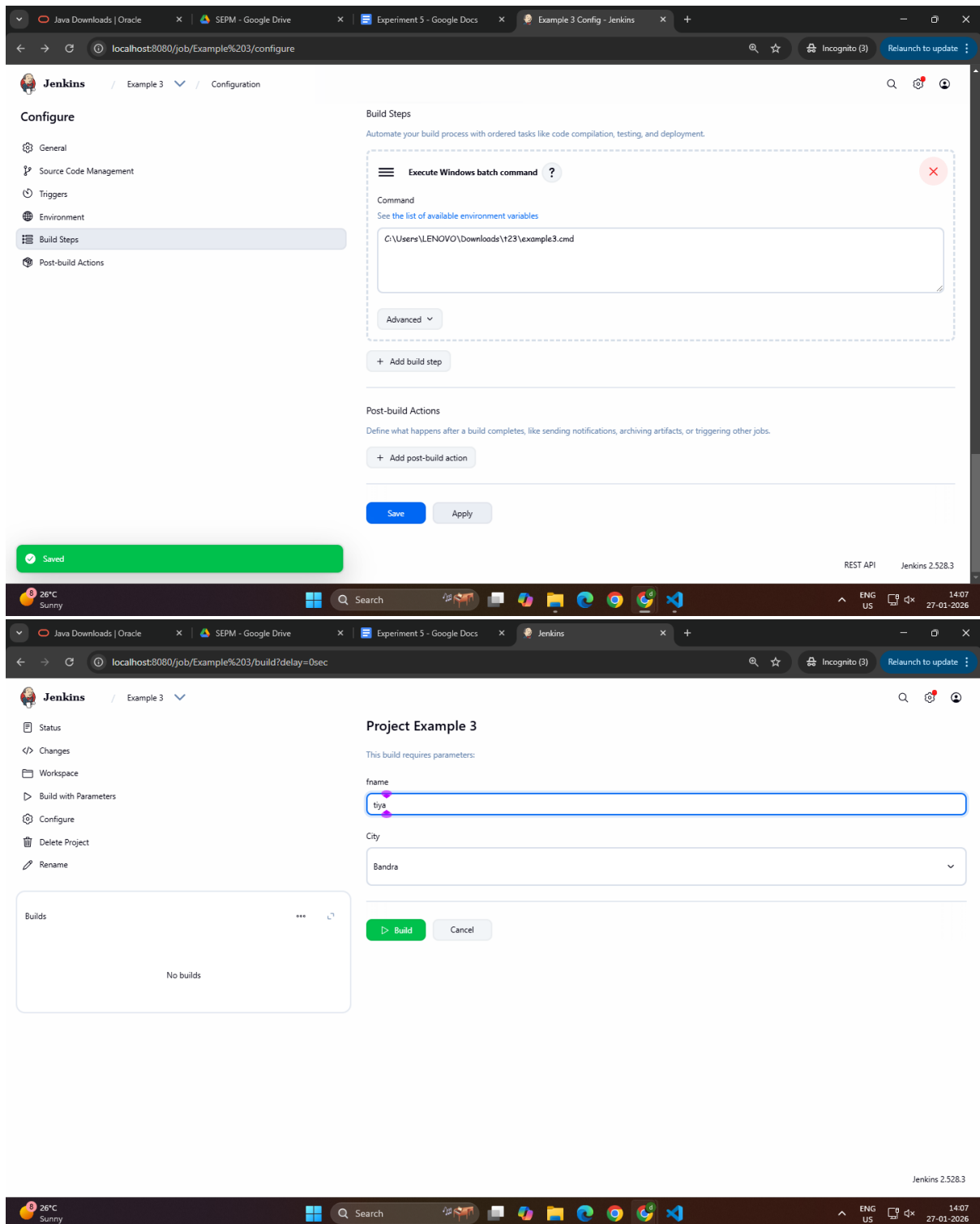
Jenkins Configuration Page:

- URL: `localhost:8080/job/Example%203/configure`
- Page Title: Jenkins / Example 3 / Configuration
- Left Sidebar: General (selected), Source Code Management, Triggers, Environment, Build Steps, Post-build Actions.
- Main Content: A blue checkmark indicates 'This project is parameterized'. Below this, there are two parameter types: 'String Parameter' and 'Choice Parameter'.
 - String Parameter:** Name: `fname`, Default Value: (empty), Description: (empty). There is a 'Trim the string' checkbox.
 - Choice Parameter:** Name: (empty).
- Buttons: 'Save' and 'Apply'.
- Notification: A green bar at the bottom left says 'Saved'.

Visual Studio Code Editor:

- File Explorer: Shows a project structure with files `example1.cmd`, `Example2.class`, `Example2.java`, `example3.cmd`, and `Main.class`.
- Editor: The file `example3.cmd` is open, showing the following content:

```
1 @echo off
2 echo "Hello %1.... Your city is %2"
```
- Terminal: A PowerShell terminal is open, showing the prompt `(base) PS C:\Users\LENOVO\Downloads\t23>`.
- Status Bar: Shows 'Ln 2, Col 29', 'Spaces: 4', 'UTF-8', 'CRLF', 'Batch', and 'Signed out'.



The image shows two screenshots of the Jenkins web interface. The top screenshot displays the 'Configure' page for a job named 'Example 3'. The left sidebar shows the configuration menu with 'Build Steps' selected. The main area is titled 'Build Steps' and contains a single step 'Execute Windows batch command'. The command field contains the text 'C:\Users\LENOVO\Downloads\t23\example3.cmd'. Below the command field is an 'Advanced' dropdown and an 'Add build step' button. The 'Post-build Actions' section is empty, with an 'Add post-build action' button. At the bottom are 'Save' and 'Apply' buttons. A green 'Saved' notification bar is visible at the bottom left. The bottom screenshot shows the 'Project Example 3' build interface. The left sidebar shows the build history menu. The main area displays the build parameters: 'fname' with the value 'tiya' and 'City' with the value 'Bandra'. Below the parameters are 'Build' and 'Cancel' buttons. The bottom status bar shows the Jenkins version 2.528.3 and the system clock.

Configure

General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute Windows batch command

Command
[See the list of available environment variables](#)

C:\Users\LENOVO\Downloads\t23\example3.cmd

Advanced

+ Add build step

Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

+ Add post-build action

Save Apply

Saved

REST API Jenkins 2.528.3

Project Example 3

This build requires parameters:

fname
tiya

City
Bandra

Build Cancel

Builds

No builds

Jenkins 2.528.3

The screenshot shows the Jenkins web interface in a browser window. The address bar displays `localhost:8080/job/Example%203/`. The page title is "Jenkins / Example 3". On the left sidebar, there are links for Status, Changes, Workspace, Build with Parameters, Configure, Delete Project, and Rename. The main content area shows the job name "Example 3" with a status icon and a "Permalinks" section containing a link for "Last build (#1), 8 ms ago". Below this, there is a "Builds" section with a search filter and a list of builds. The first build is visible, labeled "#1 2:08 PM" with a green status icon.

This screenshot shows the "Console Output" view for build #1 of the "Example 3" job. The browser address bar is `localhost:8080/job/Example%203/1/console`. The left sidebar includes links for Status, Changes, Console Output (selected), Edit Build Information, Delete build #1, Parameters, and Timings. The main content area is titled "Console Output" and shows the following text:

```
Started by user Tiya Nathwani
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\Example 3
[Example 3] $ cmd /c call C:\WINDOWS\TEMP\jenkins5651314562182551435.bat
C:\ProgramData\Jenkins\jenkins\workspace\Example 3>C:\Users\LENOVO\Downloads\t23\example3.cmd
"Hello .... Your city is "
Finished: SUCCESS
```

Buttons for "Download", "Copy", and "View as plain text" are visible at the top right of the console output area.

This screenshot shows the browser address bar with the URL `localhost:8080/job/Example 3/1/confirmDelete`. The system tray at the bottom shows the date and time as 14:08 on 27-01-2026.

3.2

The image shows a screenshot of a development environment and a web browser. The top part is an IDE (Visual Studio Code) with a Java file named `Example2.java` open. The code is as follows:

```
1 class Example2 {
2     public static void main(String args[]) {
3         int i;
4         int num = Integer.parseInt(args[0]);
5         for(i = 1; i < 11; i++)
6             System.out.println(i + "Hello World!");
7     }
8 }
```

The bottom part of the image shows a web browser displaying the Jenkins "New Item" page. The item name is set to "Example4". The item type is selected as "Pipeline". The description for "Pipeline" is: "Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents." The "OK" button is visible at the bottom of the form.

The image displays two screenshots of the Jenkins web interface, showing the configuration of a job named 'Example4'.

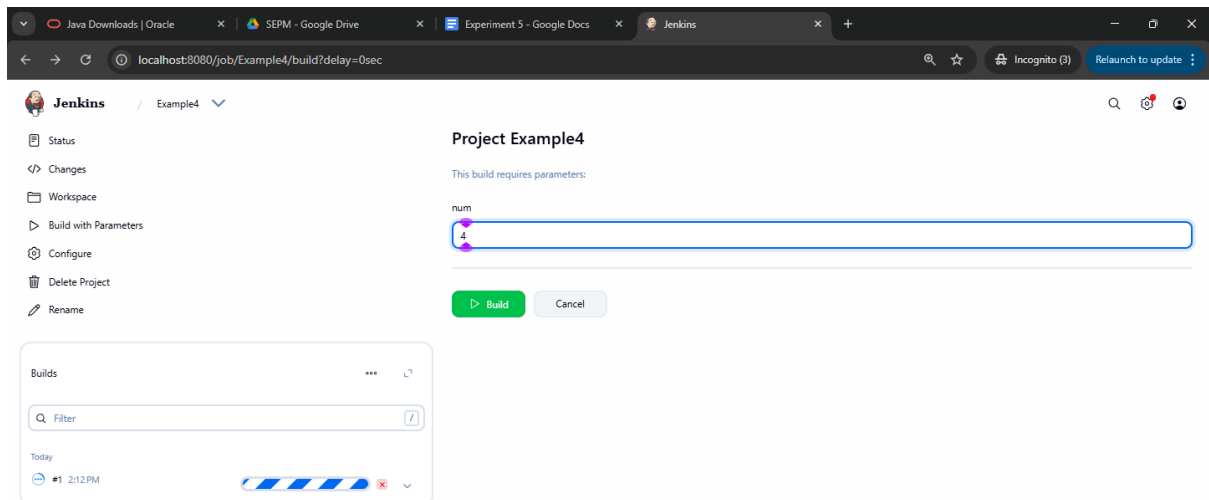
Top Screenshot: String Parameter Configuration

- The browser address bar shows `localhost:8080/job/Example4/configure`.
- The Jenkins configuration page for 'Example4' is shown, with the 'General' tab selected.
- The 'This project is parameterized' checkbox is checked.
- A 'String Parameter' is being configured with the name 'num'.
- The 'Default Value' field is empty.
- The 'Description' field is empty.
- The 'Trim the string' checkbox is unchecked.
- The 'Add Parameter' button is visible at the bottom.

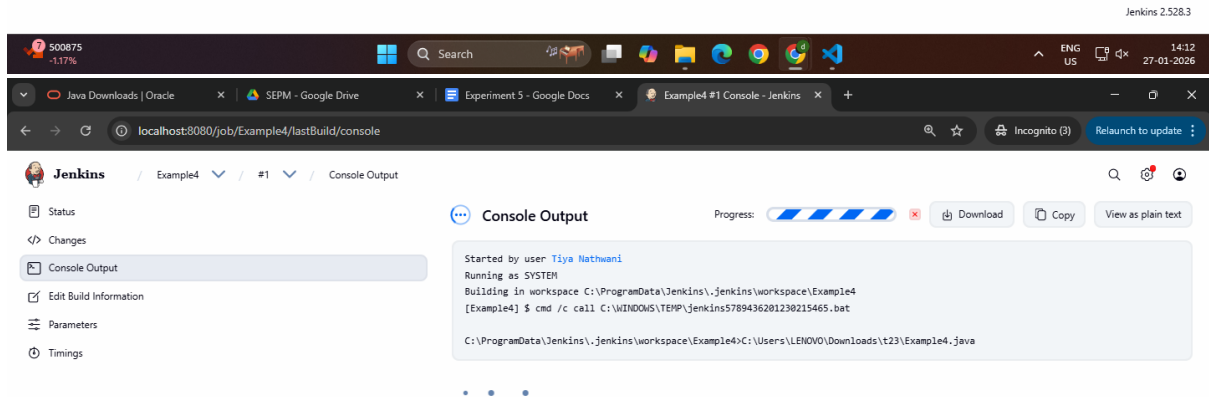
Bottom Screenshot: Build Steps Configuration

- The browser address bar shows `localhost:8080/job/Example4/configure`.
- The Jenkins configuration page for 'Example4' is shown, with the 'Build Steps' tab selected.
- The 'Build Steps' section is titled 'Automate your build process with ordered tasks like code compilation, testing, and deployment.'
- An 'Execute Windows batch command' step is configured with the command `C:\Users\LENOVO\Downloads\123\Example4.java %num%`.
- The 'Advanced' dropdown is visible.
- The 'Add build step' button is visible.
- The 'Post-build Actions' section is titled 'Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.'
- The 'Add post-build action' button is visible.
- The 'Save' and 'Apply' buttons are at the bottom.

A green 'Saved' notification bar is visible at the bottom of the second screenshot.



The screenshot shows the Jenkins web interface for a project named 'Example4'. The browser address bar indicates the URL is `localhost:8080/job/Example4/build?delay=0sec`. On the left sidebar, navigation links include Status, Changes, Workspace, Build with Parameters, Configure, Delete Project, and Rename. The main content area is titled 'Project Example4' and states 'This build requires parameters:'. A parameter field labeled 'num' is present with a value of '1'. Below the field are 'Build' and 'Cancel' buttons. A 'Builds' section on the left shows a list of builds with a filter input and a table containing one entry: '#1 2:12 PM' with a progress bar. The top of the page shows the Jenkins logo and the project name 'Example4'.



The screenshot shows the Jenkins web interface for the 'Console Output' of build '#1' of 'Project Example4'. The browser address bar indicates the URL is `localhost:8080/job/Example4/lastBuild/console`. The left sidebar includes links for Status, Changes, Console Output (which is selected), Edit Build Information, Parameters, and Timings. The main content area is titled 'Console Output' and shows a progress bar. The console output text is as follows:
Started by user [Tiya Nathwani](#)
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\Example4
[Example4] \$ cmd /c call C:\WINDOWS\TEMP\jenkins5789436281238215465.bat
C:\ProgramData\Jenkins\jenkins\workspace\Example4>C:\Users\LENOVO\Downloads\t23\Example4.java
The top of the page shows the Jenkins logo and the project name 'Example4'.